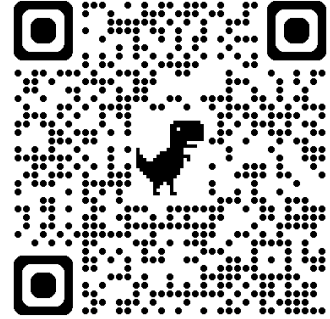
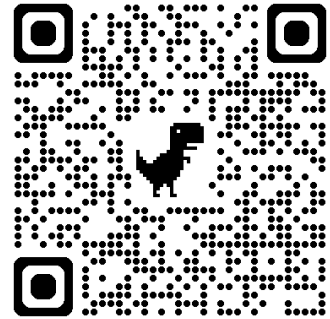


- Session Objectives:
 1. Understand the basics of sleep physiology
 2. Recognize, diagnose and manage patients with sleep disorders, including use of the polysomnogram for evaluation and when to obtain outside consultation
 3. Understand the importance of sleep wellness to optimize provider cognition and patient safety
- Connections
 - ScholarRx: Sleep physiology; Sleep Disorders: QR→
 - Lecture on Pharmacology of Sedatives/Hypnotics (PPOM1)
 - Lecture on Sleep Physiology and Sleep Apnea (PPOM2)



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Sleep Wake Cycle

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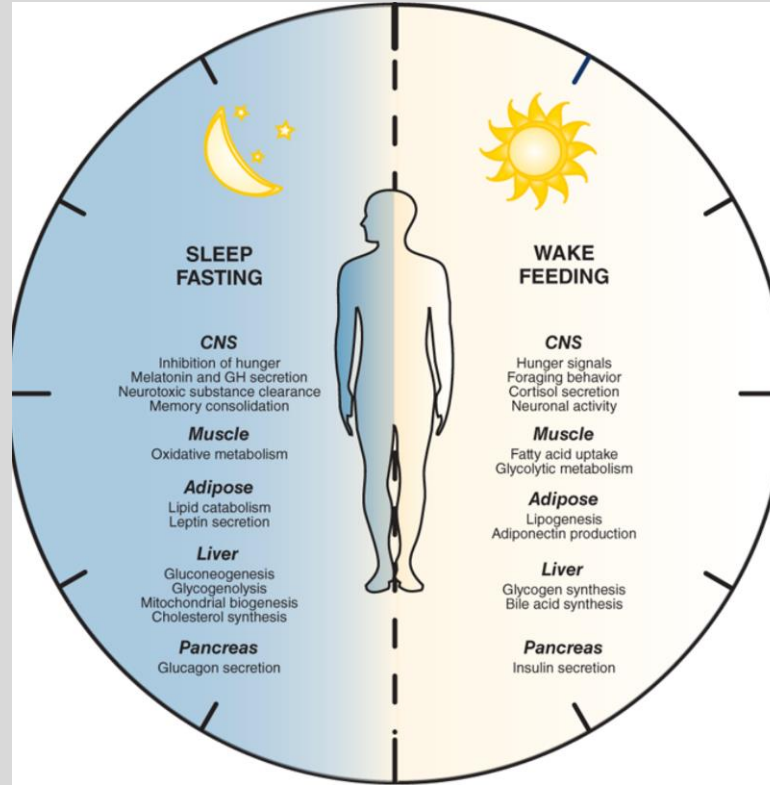
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Medicine

Fun Fact 1!

Circadian rhythms are present in almost all living things.

Fun Fact 2!

Circadian rhythms result in physical, mental and behavioral changes.

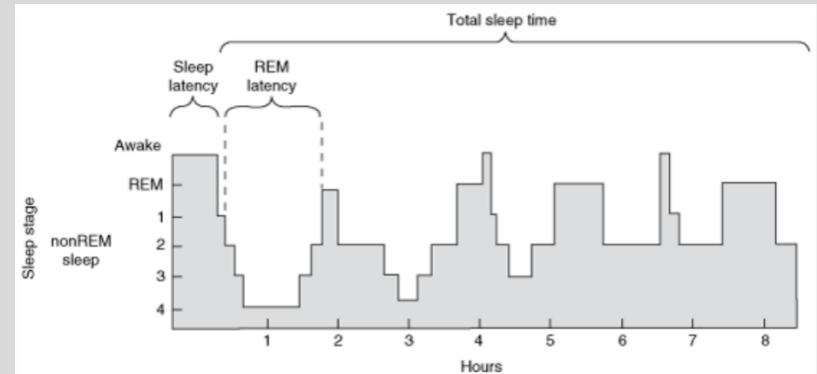
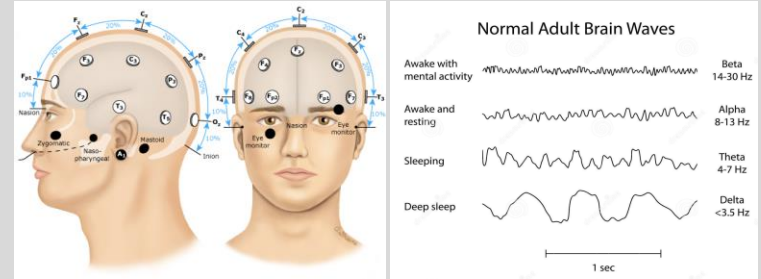


Fun Fact 3!

Our body's clock cycle without the influence of the day-night cycle is actually 25 hours.

Measuring Sleep: The Polysomnogram

- Overnight sleep study
- EEG, EOG, EMG, ECG, airflow, respiratory effort, O2 saturation, limb movements, etc.
- Usually performed in sleep lab (Neurology, Pulmonary)
- Gold standard to diagnosis sleep apnea, narcolepsy, parasomnias, RLS



Source: Ebert MH, Loosen PT, Nurcombe B, Leckman JF:
Current Diagnosis and Treatment: Psychiatry, 2nd Edition:
<http://www.accessmedicine.com>

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EEG = electroencephalogram, EOG = electrooculogram, EMG = electromyogram, ECG = electrocardiogram, RLS = restless legs syndrome, REM = rapid eye movements

Adapted from: Jasper HH. Report of the committee on methods of clinical examination in electroencephalography: 1957. Electroencephalogr Clin Neurophysiol 1958; 10:370.

Categorization of Sleep Disorders

- **Dyssomnias** – disorders of sleep-wake time and/or quality
 - Primary (intrinsic)
 - Primary insomnia
 - Sleep-related movement disorders, e.g., restless legs syndrome
 - Central disorders of hypersomnolence, e.g., narcolepsy
 - Secondary (extrinsic)
 - Secondary insomnia, e.g., depression, anxiety
 - Circadian rhythm sleep-wake disorders, e.g., jet-lag
 - Obstructive sleep apnea, congestive heart failure, GERD, acute illness, etc.
- **Parasomnias** – physical behaviors (e.g., sleepwalking) or experiences (e.g., sleep terrors) that occur during or around sleep

General Treatment of Insomnia

- Treat underlying conditions or discontinue substances that promote insomnia
- Regular exercise and nutritious diet
- Sleep Hygiene
- Nonbenzodiazepine hypnotics
 - Zolpidem, Zaleplon, esZopiclone (ZZZ...)

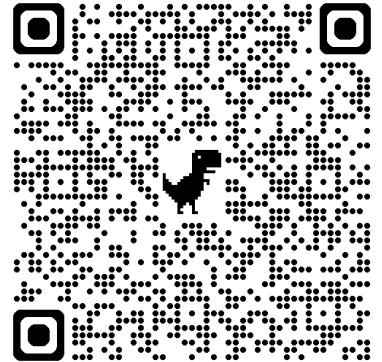
Specific Treatment of Insomnia

- 1st Line: sleep hygiene
- 2nd: pharmacologic: ZZZ..., Melatonin receptor agonists
- 3rd: benzodiazepines* – disrupts sleep cycle, addiction
- 4th: antipsychotics, antidepressants, antihistamines – last resorts due to high side effect profile
- Above may be different for specific sleep disorders, e.g., narcolepsy

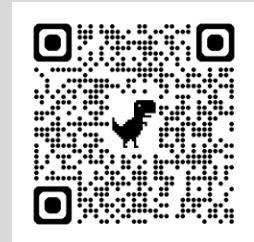
*In the past, benzodiazepines were so commonly prescribed 1st or 2nd line for insomnia that you may see them still prescribed earlier than the above scheme, a prescriber habit that should be updated.

Consequences of Sleep Deprivation

- Fatal familial insomnia, a rare prion disease, results in progressive insomnia and eventual death
- A few days without sleep causes progressive slowed, altered cognition and motor control; hallucinations; paranoia; memory loss; and psychosis
- Guinness record for longest awake is 18 days, 21 hours and 40 minutes without sleep
- Guinness no longer monitors or records sleep deprivation due to the negative health effects
- The risk of motor vehicle accidents increases significantly with sleep deprivation



GuinnessWorldRecords.com



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